

⚡ Visualization Tool for Electric Vehicle Charge and Range Analysis

📌 Project Overview

The Visualization Tool for Electric Vehicle Charge and Range Analysis is an interactive data visualization project developed to analyze electric vehicle (EV) market trends, vehicle specifications, charging infrastructure, battery range, pricing, efficiency, body styles, and powertrain distribution.

The project integrates multiple electric vehicle datasets using MySQL for data management and SQL for data analysis. Interactive dashboards and stories are created using Tableau Desktop, published through Tableau Public, and embedded into a Flask web application for an intuitive and user-friendly experience.

🎯 Project Objectives

The main objectives of this project are:

- Analyze electric vehicle charging infrastructure.
- Compare EV brands and vehicle models.
- Visualize vehicle price distribution.
- Analyze driving range and vehicle performance.
- Compare energy efficiency across brands.
- Study body style and powertrain distribution.
- Provide interactive dashboards for data-driven decision-making.

🛠 Technologies Used

Database

- MySQL Server 8.0
- MySQL Workbench

Data Analysis

- SQL

Data Visualization

- Tableau Public

Web Integration

- Python Flask
- HTML5
- CSS3
- Bootstrap

- JavaScript

Development Tools

- Visual Studio Code
- Git
- GitHub

📁 Dataset Used

The project integrates four electric vehicle datasets:

- ElectricCarData_Clean.csv
- EVIndia.csv
- Electric Vehicle Charging Station List.csv
- CheapestElectricCars-EVDatabase.csv

⚙️ Project Workflow

```text

EV Datasets



Data Cleaning & Preprocessing



MySQL Database



SQL Analysis



Tableau Desktop



Interactive Dashboard & Story



Tableau Public



Flask Web Application



End User

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Tableau Visualizations

The project includes the following visualizations:

- Charging Stations by Region and Type
- EV Charging Station Map
- Different Electric Vehicle Cars
- Top Speed by Brand
- Price Comparison of EV Cars
- Top 10 Efficient EV Brands
- Brands According to Body Style
- Powertrain Treemap
- Different EV Brands (Global KPI)
- Different EV Brands in India (KPI)

Dashboard Features

- Interactive KPI Cards
- Dynamic Filters
- Dashboard Actions
- EV Charging Infrastructure Analysis
- Vehicle Price Comparison
- Brand-wise Performance Analysis
- Battery Range Analysis
- Efficiency Comparison
- Interactive Geographic Maps
- Body Style Distribution
- Powertrain Analysis

Tableau Story

The Tableau Story presents EV analytics through multiple scenes, including:

- Dashboard Overview
- EV Charging Infrastructure Analysis
- Vehicle Brand Comparison
- Price and Performance Analysis
- Efficiency Analysis
- Key Insights and Conclusions

Web Integration

A Flask web application was developed to integrate the Tableau Dashboard and Tableau Story into a single interface.

Features

- Responsive User Interface
- Embedded Tableau Dashboard
- Embedded Tableau Story
- Easy Navigation
- Interactive User Experience

📁 Project Structure

```
```text
Visualization-Tool-for-Electric-Vehicle-Charge-and-Range-Analysis/
├── 1.Ideation_Phase/
├── 2.Requirement_Analysis/
├── 3.Project_Design_Phase/
├── 4.Project_Planning_Phase/
├── 5.Project_Development_Phase/
│ ├── datasets/
│ ├── SQL_Queries/
│ ├── Dashboard_Report.pdf
│ └── Story_Report.pdf
├── 6.Performance_Testing/
├── 7.Project_Documentation/
├── templates/
│ └── index.html
├── app.py
├── requirements.txt
└── README.md
```
```

🔗 Project Links

📊 Tableau Dashboard

[https://public.tableau.com/views/ElectricCarsAnalyticsDashboard_17831451243460/Dashboar
d1](https://public.tableau.com/views/ElectricCarsAnalyticsDashboard_17831451243460/Dashboar
d1)

📖 Tableau Story

https://public.tableau.com/views/StoryofElectricCarsinIndia_17832669081640/StoryofElectricCarsinIndia

📌 Results

- Developed an interactive EV analytics platform.
- Integrated MySQL, SQL, Tableau, and Flask.
- Published dashboards and stories on Tableau Public.
- Embedded visualizations into a Flask web application.
- Enabled data-driven analysis of EV charging infrastructure, pricing, range, and performance.

🚀 Future Enhancements

- Real-time EV dataset integration
- Live charging station availability
- AI-based EV recommendation system
- Machine Learning for EV trend prediction
- Battery health and charging time analysis
- Cloud deployment
- Mobile application support
- Global EV market analytics

🧑 Developed By

Team Members

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